

StructRL: Recovering Dynamic Programming Structure from Learning Dynamics in Distributional Reinforcement Learning

Ivo Nowak

HAW-Hamburg

Email: ivo.nowak@haw-hamburg.de

Abstract—Reinforcement learning is typically treated as a uniform optimization process that does not explicitly exploit global structure, leading to slow and inefficient information propagation. In contrast, dynamic programming achieves high efficiency by leveraging structured propagation of value information.

In this paper, we argue that such structure is implicitly encoded in its learning dynamics. Focusing on distributional reinforcement learning, we show that the temporal evolution of return variance reveals a propagation process across the state space.

We introduce a temporal learning indicator $t^*(s)$ that captures when a state is most strongly affected by propagated updates. Empirically, this signal induces an ordering over states consistent with dynamic programming-style propagation, where information spreads from terminal regions outward.

Based on this insight, we propose StructRL, a framework that exploits this emergent structure to guide exploration and sampling. In a deterministic single-goal gridworld setting, this leads to substantially faster and more stable convergence compared to a standard baseline. We further validate StructRL on Taxi 10×10 environment, a standard sparse-reward benchmark, achieving 169 ± 25 successful episodes out of 300 versus 13 ± 12 for a C51 baseline — a twelve-fold improvement, evaluated over 3 independent seeds.

Our results support a new perspective: reinforcement learning can be understood as a structured propagation process, and learning dynamics themselves provide actionable information to control it. This opens a pathway toward more sample- and energy-efficient reinforcement learning.

I. INTRODUCTION

REINFORCEMENT learning (RL) plays a central role in goal-directed artificial intelligence, enabling agents to learn behavior that optimizes long-term objectives through interaction with an environment [4], see also “The era of experience” [7]. However, modern RL methods often suffer from slow learning, high sample complexity, and instability, partly because they treat learning as a uniform optimization process.

Learning is driven by rewards and temporal-difference errors, while global structure in the environment remains largely unused. As a result, information propagates slowly and often inefficiently across the state space. Techniques such as prioritized experience replay [3] improve efficiency but remain based on local signals rather than global propagation structure.

In contrast, classical approaches such as dynamic programming (DP) explicitly exploit structure [5]. In shortest-path

problems, value updates follow an ordering induced by the distance-to-goal $d(s)$, which measures the minimal number of steps from a state to the goal. This enables efficient backpropagation of information from the goal to all other states.

This raises the question:

Can such distance structure be recovered and exploited from the learning dynamics of model-free methods?

In this paper, we investigate whether such structure can be recovered directly from the learning dynamics of model-free methods, in particular distributional reinforcement learning [1], [2].

We analyze the temporal evolution of return distributions and introduce a temporal learning indicator $t^*(s)$ that captures when a state is most strongly affected by propagated updates.

Empirically, we observe that $t^*(s)$ induces an ordering over states that is consistent with a dynamic programming-style propagation of information [5]. States that are structurally closer to the goal tend to become active earlier, while more distant states are updated later. This results in a layered temporal organization of learning.

Building on this observation, we introduce StructRL, a framework that exploits this recovered structure to guide learning. Importantly, StructRL does not rely on an explicit model or prior knowledge of the environment. Instead, it uses $t^*(s)$ to identify a small set of anchor states (see Sec. III-E), from which a structural distance function is constructed. The transition graph used to compute this distance function is built entirely from observed transitions stored in the replay buffer, without access to a simulator or environment model — making StructRL model-free in the standard sense.

We then demonstrate, in a minimal proof-of-concept setting, that this structure can be used to bias exploration and sampling, leading to more efficient propagation of information. Our goal is not to present a fully optimized algorithm, but to show that learning dynamics themselves provide actionable structural signals that can be exploited.

A. Contributions

- **A new perspective on reinforcement learning.** We argue that learning in distributional reinforcement learning can be interpreted as a structured propagation process

across the state space, rather than a uniform optimization procedure.

- **Temporal learning indicator.** We introduce $t^*(s)$, a signal derived from the evolution of return variance, which captures when states are most affected by propagated updates.
- **Recovery of propagation structure.** We show that $t^*(s)$ induces a temporal ordering over states consistent with dynamic programming-style propagation, without requiring an explicit model.
- **StructRL framework.** We propose a simple mechanism that exploits this structure to guide exploration and replay via a derived distance function.
- **Empirical support.** In a deterministic single-goal grid-world setting, we demonstrate that this structure-aware bias leads to faster and more stable learning, indicating that the recovered structure is actionable. On Taxi 10×10 environment, StructRL achieves 169 ± 25 goals out of 300 episodes — a twelve-fold improvement over a C51 baseline, evaluated over 3 seeds.

II. RELATED WORK

Prioritized experience replay [3] biases replay sampling toward transitions with large TD errors. StructRL is different: rather than prioritizing by local error magnitude, it prioritizes by structural progress toward anchor states.

Prioritized sweeping [8] propagates value updates backward from recently changed states using an explicit model. StructRL achieves a similar effect without a model, using the observed replay graph as a proxy for the transition structure.

Count-based and curiosity-driven exploration methods [9], [10] encourage visiting novel states. StructRL differs by directing exploration toward structurally relevant states rather than merely unvisited ones.

The options framework [11] defines temporally extended actions toward subgoals. StructRL identifies anchor states analogous to subgoals but without requiring pre-specified option policies.

To our knowledge, no prior work extracts DP-style propagation structure directly from the variance dynamics of a distributional RL agent.

III. TEMPORAL STRUCTURE IN DISTRIBUTIONAL RL

A. Return Variance

In distributional reinforcement learning, the return is modeled as a random variable $Z(s, a)$ rather than a scalar value [1], [2]. This provides access to richer learning signals, in particular information about the dynamics of learning.

We consider the return variance

$$\sigma(s) = \sqrt{\text{Var}[Z(s)]}, \quad (1)$$

and denote by $\sigma_t(s)$ its value at training step t . **Initialisation.** The C51 output layer is initialised with zero weights and a positive bias on the centre atom, concentrating the initial return distribution on a single value so that $\sigma_0(s) \approx 0$ for all s . With

this initialisation, $\sigma_t(s)$ remains near zero until the variance wave from a reward source first reaches s .

Exact variance recursion. Under the C51 Bellman update with learning rate α , the return variance satisfies:

$$\sigma_{t+1}^2(s) = (1-\alpha)^2 \sigma_t^2(s) + \alpha^2 \sigma_t^2(s') + \alpha(1-\alpha) \delta_t^2(s) \quad (2)$$

where $\delta_t(s)$ is the mean TD error. Due to the recursive structure of the distributional Bellman updates,

$$Z(s) \leftarrow \mathcal{T}Z(s'), \quad (3)$$

changes in the return distribution propagate from successor states to predecessors over time.

As a result, $\sigma_t(s)$ does not evolve uniformly. Instead, it typically exhibits a transient increase when new information reaches a state, followed by stabilization.

This temporal behavior reflects how learning propagates through the state space, rather than merely capturing uncertainty magnitude.

B. Temporal Evolution of Learning

To analyze learning dynamics, we consider the temporal evolution of the variance signal $\sigma_t(s)$. We study how this signal changes over time. In particular, we are interested in identifying when learning at a given state is most active.

C. Temporal Learning Indicator

We define a temporal learning indicator $t^*(s)$ based on the strongest positive change in the variance signal:

$$t^*(s) = \arg \max_t \max (\sigma_{t+1}(s) - \sigma_t(s), 0). \quad (4)$$

This quantity captures the time at which the variance at state s exhibits its strongest positive change. Intuitively, it identifies when propagated information reaches s most strongly through the recursive Bellman updates. In this sense, $t^*(s)$ is not a measure of uncertainty magnitude, but a temporal marker of information arrival.

In a *single-reward* setting and $\sigma_0(s) \approx 0$, equation (2) predicts two phases for $\sigma_t(s)$: (i) *flat*: near zero while s has not yet been connected to any reward source; (ii) *rise*: a sharp increase at $t^*(s)$ when the variance wave first arrives, after which $\sigma_t(s)$ stabilises at a nonzero level determined by the variance propagated from successor states. In multi-reward settings such as Taxi 10×10 environment, variance waves from different reward sources interfere in the shared network, keeping $\sigma_t(s)$ elevated and making the flat phase less well-defined. Separate networks per reward source would give cleaner phase separation and is a direction for future work.

D. Interpretation

We interpret $t^*(s)$ as a temporal signal indicating when learning propagates through a state. Empirically, we observe that this signal induces an ordering over states that is consistent with a structured propagation of information.

Importantly, this structure is not uniform. States are organized into temporal layers, and learning progresses along transitions

connecting these layers. This suggests that learning dynamics implicitly define a propagation process over the state space.

In the following sections, we show how this temporal structure can be exploited to guide both exploration and sampling.

E. Anchor States and Distance Function

A key step in StructRL is the identification of a small set of anchor states S_0 , which define the origin of the recovered propagation structure.

We estimate S_0 from the temporal learning signal $t^*(s)$ by selecting states that become active early during training, i.e., states with low $t^*(s)$. These states typically correspond to regions close to the goal or to structurally favorable parts of the state space.

To improve robustness under limited data, we combine this temporal signal with a simple stability criterion: states whose greedy action remains stable in later training stages are preferred, as they indicate partially converged value estimates.

This results in a sparse set of early-active and locally stable states that serve as reliable seeds for constructing the structural distance function $d(s)$.

Distance function. The transition graph used to compute $d(s)$ is the *replay graph* $\mathcal{G} = (V, E)$: a directed graph where V contains all states observed in the replay buffer, and $(s, s') \in E$ if the transition $s \rightarrow s'$ has been observed at least $n_{\min}$ times. No simulator or environment model is required. The structural distance is then:

$$d(s) = \min_{g \in S_0} \text{BFS-dist}_{\mathcal{G}}(s, g). \quad (5)$$

This is analogous to model-based prioritized sweeping [8] but requires no explicit model.

IV. TEMPORAL FRONTIERS OF LEARNING

The temporal learning indicator $t^*(s)$ reveals more than isolated learning events. It induces a structured organization of states according to when learning becomes most active.

A. Temporal Frontiers

The temporal learning indicator $t^*(s)$ induces a structured organization of states. States closer to the goal tend to become active earlier, while more distant states become active later, forming temporal layers.

Learning can therefore be interpreted as a propagation process across these layers, rather than a uniform update over the state space.

B. Sparse Transition Structure

Importantly, we observe that learning concentrates on a sparse set of transitions that connect different temporal frontiers. In particular, only a small subset of transitions ($s \rightarrow s'$) satisfy both:

- alignment in temporal structure, i.e., $|t^*(s) - t^*(s')|$ is small,
- directional progress, i.e., the transition moves toward structurally favorable regions.

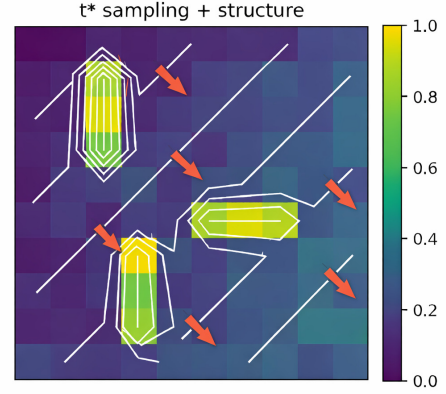


Fig. 1. Illustration of the StructRL mechanism. The temporal learning signal $t^*(s)$ is used to identify a small set of anchor states, from which a structural distance function $d(s)$ is constructed. Learning is then biased toward transitions that reduce this distance (red arrows), resulting in a structured propagation of information across the state space.

These transitions form a sparse backbone along which learning propagates. Rather than uniformly covering the state space, the learning dynamics focus on a limited number of key pathways.

C. Connection to Dynamic Programming

This behavior is similar to dynamic programming, where information propagates in a structured way from favorable states to the rest of the state space. However, in contrast to classical dynamic programming, this structure is not imposed explicitly. Instead, it emerges from the interaction between learning updates and the environment.

V. FROM STRUCTURE TO STRUCTRL

The temporal structure induced by $t^*(s)$ suggests that learning follows a structured propagation process across the state space. We now investigate whether this recovered structure can be exploited to guide learning.

Within the StructRL framework, we consider a simple mechanism that uses this recovered structure to bias both exploration and replay. The goal is to test whether aligning updates with this structure improves the propagation of information.

Importantly, $t^*(s)$ is not used directly for control. Instead, it serves only to identify a small set of anchor states S_0 , corresponding to early-active regions of the state space. From this anchor set, we construct a structural distance function $d(s)$, defined as the shortest-path distance to S_0 under the environment dynamics.

This induces a dynamic programming-style structure over the state space [5], where information can propagate from the anchor states to more distant regions.

A. Core Idea of StructRL

Fig. 1 illustrates the core idea of StructRL. The key insight is that learning dynamics reveal a propagation structure over

the state space. Anchor states define the origin of this structure, and the induced distance $d(s)$ organizes states into layers.

Instead of updating transitions uniformly, StructRL prioritizes transitions that reduce $d(s)$, leading to a structured propagation of information.

B. Frontier-Aligned Exploration

We modify exploration by biasing action selection toward transitions that are aligned with the recovered propagation structure.

a) *Directional Signal*: We favor transitions that reduce the structural distance, i.e., $d(s') < d(s)$, implemented via a soft preference:

$$\text{direction}(s, s') = \exp(\lambda(d(s) - d(s'))), \quad (6)$$

where λ controls the strength of the bias.

b) *Exploration Component*: This directional signal is combined with stochastic exploration to maintain sufficient state-space coverage [4].

c) *Action Selection*: Actions are sampled proportionally to their directional score, combined with stochastic exploration.

C. Structure-Aware Replay

Given a transition (s, a, s') , we assign a replay score based on directional progress:

$$\text{score}(s, s') = \tanh(\alpha(d(s) - d(s'))), \quad (7)$$

where α controls the sharpness of the preference. Transitions that reduce the structural distance are sampled more frequently.

D. Combined Effect

The combination of frontier-aligned exploration and structure-aware replay induces a learning dynamic in which information propagates along a sparse set of transitions, similar to dynamic programming but without an explicit model.

VI. PRELIMINARY EMPIRICAL EVIDENCE

A. Single-Goal Gridworld Environment

Our first experiment is conducted in a deterministic 10×10 single-goal gridworld using C51 [1]. Each transition yields reward -1 ; episodes terminate at the goal (upper left corner). The optimal value function is $J^*(s) = -d(s)$.

B. Structural Signals

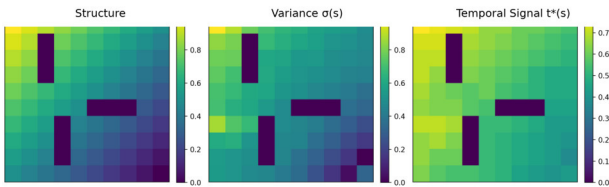


Fig. 2. Single-goal gridworld: Normalized heatmaps of structural distance $d(s)$, return variance $\sigma(s)$, and temporal learning indicator $t^*(s)$.

Fig. 2 shows three complementary signals. A key observation is that $t^*(s)$ exhibits a clear spatial structure consistent with BFS distance to the goal: states closer to the goal become active earlier. This indicates that learning progresses gradually across the state space.

C. Temporal Ordering and Structure

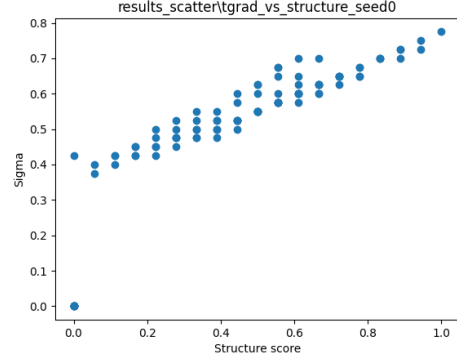


Fig. 3. Single-goal gridworld: Relationship between structural distance $d(s)$ and temporal learning indicator $t^*(s)$.

Fig. 3 shows the relationship between $t^*(s)$ and the true distance-to-goal $d(s)$. While the relationship is not perfectly linear, a clear trend emerges: states with similar structural depth tend to be updated at similar training stages, consistent with dynamic programming-style propagation [5].

D. Structure-Induced Learning Dynamics

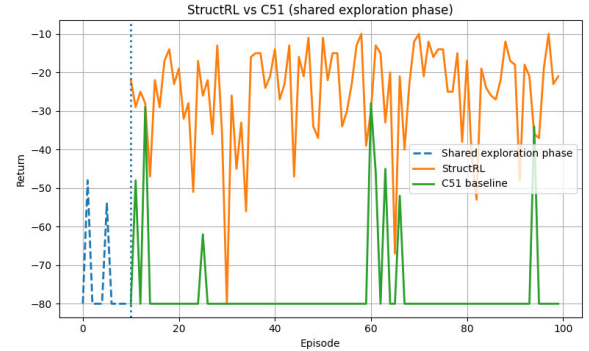


Fig. 4. Single-goal gridworld: StructRL vs. C51 [1]. StructRL improves faster after exploration due to structure-aware propagation.

Fig. 4 shows the learning curves. The baseline agent struggles with slow and unstable learning. StructRL rapidly improves after the initial exploration phase, concentrating updates on transitions that align with the recovered structure.

E. Summary

- $t^*(s)$ reveals structured propagation consistent with distance-to-goal in single-reward settings.
- StructRL exploits this structure by biasing transitions, leading to faster and more stable learning.

VII. MULTI-STAGE TAXI ENVIRONMENT

In contrast to the single-goal gridworld, Taxi requires the agent to solve two sequential subproblems, passenger pickup and correct dropoff, and therefore contains multiple interacting reward sources.

Setup. Our Taxi 10×10 environment is a multi-stage, sparse-reward variant of the Taxi domain introduced by Dietterich [12] with four stations $R(0, 0)$, $G(0, 9)$, $Y(9, 0)$, $B(9, 9)$. State: (row, col, pass_loc, dest) — 2000 states. Rewards: pickup +5, dropoff +20, wrong action -10, step -1. Success ($\text{done}=\text{True}$) is defined as the passenger being correctly dropped off. 300 episodes, 500 steps per episode, 3 seeds.

Results. Table I shows the main result. StructRL achieves 169 ± 25 successful episodes versus 13 ± 12 for the C51 baseline — a twelve-fold improvement. The gap between pickups (221 ± 17) and completed goals (169 ± 25) confirms that the main difficulty lies in the second reward source (dropoff). Anchor states are identified automatically near the pickup and dropoff stations without any environment-specific code.

TABLE I
TAXI 10×10 : GOALS AND PICKUPS OUT OF 300 EPISODES (MEAN $\pm$ STD, 3 SEEDS).

Method	Goals	Pickups
C51 baseline	13 ± 12	48 ± 20
StructRL	169 ± 25	221 ± 17
Improvement	+1200%	+360%

Fig. 5 shows the $t^*(s)$ heatmap for Phase 0, where dark cells indicate early learning consistent with proximity to reward sources.

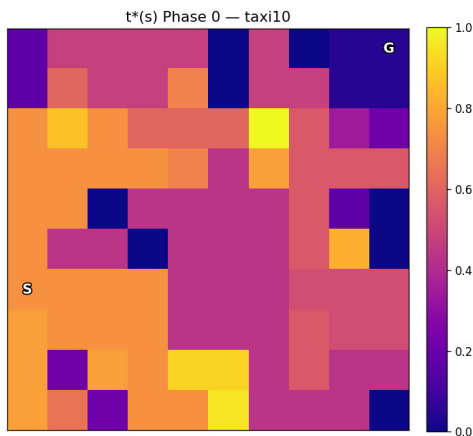


Fig. 5. Taxi 10×10 : $t^*(s)$ heatmap for Phase 0. Dark cells indicate early learning and structural proximity to a reward source. G marks one of the designated Taxi stations.

VIII. DISCUSSION

A. Learning as Structured Propagation

A central observation is that learning does not evolve uniformly. Instead, updates appear in a temporally ordered manner, progressing from structurally favorable regions outward. The temporal signal $t^*(s)$ makes this process observable.

B. Relation to Dynamic Programming

The ordering induced by $t^*(s)$ is consistent with DP propagation patterns [5] but emerges without an explicit model — it is a property of the learning process, not of explicit planning.

C. Effect of Structure-Aware Bias

Introducing a structure-aware bias concentrates updates on transitions aligned with the recovered structure, leading to faster and more consistent discovery of goal-reaching trajectories without modifying the underlying learning algorithm.

D. Why StructRL Outperforms C51

C51 allocates equal learning capacity to all states. In Taxi 10×10 environment, many structurally important states are rarely visited because rewards are sparse. StructRL concentrates both exploration and replay on states along structural paths to anchor states, approximating the reverse-topological ordering that makes DP efficient. Once anchors are correctly placed, BFS on the replay graph computes reliable structural distances, and the exploration bias guides the agent efficiently.

E. Limitations

Several limitations remain. First, the $t^*(s) \propto d(s)$ relationship is empirical and holds cleanly only in single-reward settings. Second, the approach relies on sufficient exploration. Third, the current scoring function uses hand-designed signal combinations. Fourth, with a single global network, variance waves from different reward sources mix, reducing the precision of $t^*(s)$ in multi-reward settings. Separate networks per reward source would give cleaner signals and is a direction for future work.

F. Extension to Continuous Control

StructRL can be combined with C-DSAC [6] for continuous domains, where analogous variance signals may emerge from high-dimensional control tasks.

IX. CONCLUSION

We introduced $t^*(s)$, a temporal learning indicator that captures when a state undergoes its strongest update, and proposed StructRL, a framework that exploits the recovered structure to bias exploration and replay. The key contribution is a new perspective: learning dynamics contain actionable structural information that can transform uniform learning into structured propagation. Validation on the more challenging Taxi 10×10 environment provides further preliminary evidence: 169 ± 25 goals versus 13 ± 12 for C51 over 3 seeds, a twelve-fold improvement.

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HYPERPARAMETERS

TABLE II
STRUCTRL HYPERPARAMETERS (ALL EXPERIMENTS).

Parameter	Value	Description
α	0.001	C51 learning rate
γ	0.99	Discount factor
ε_0	1.0	Initial exploration
$\varepsilon_{\min}$	0.05	Final exploration
Atoms	51	C51 atoms
Buffer size	100000	Replay buffer
Batch size	64	Training batch
$n_{\min}$	1	Min edge count
λ	1.0	Exploration bias
$n_{\max}$	4	Max anchor states
$d_{\min}$	2	Min anchor distance
Update frequency	5 ep	Structure update

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